

Fintech Lending Platform - Architecture

Overview

The Fintech Lending Platform is being built as a modern digital lending platform targeting the Indian market.

Initial loan products:

- Personal Loans
- MSME Loans

Future capabilities:

- Lender Marketplace
 - P2P Lending
 - AI-Based Underwriting
 - Risk Scoring Engine
-

Technology Stack

Frontend

- Flutter
- Single codebase
- Web
- Android
- iOS

Backend

- FastAPI
- Python 3.11

Database

- PostgreSQL

ORM

- SQLAlchemy

API Documentation

- Swagger/OpenAPI

Cloud (Future)

- Google Cloud Platform (GCP)
 - Cloud SQL
 - Cloud Run
 - Cloud Storage
-

Current Architecture

Flutter (Web + Mobile)

↓

FastAPI

↓

SQLAlchemy

↓

PostgreSQL

Repository Structure

fintech-lending-platform/

├── backend/

├── frontend/

├── docs/

└── infra/

Backend Structure

backend/

├── app/

```
| |—— core/
| |—— db/
| |—— models/
| |—— schemas/
| |—— routers/
| |—— services/
|
|—— logs/
|—— venv/
|—— start_backend.sh
|—— stop_backend.sh
|—— requirements.txt
```

Design Principles

1. Modular Monolith
 2. API First
 3. Database First Design
 4. Security by Default
 5. Cloud Portable
 6. Documentation Driven
-

Current Status

Completed:

- Backend setup
- Frontend setup
- PostgreSQL integration
- SQLAlchemy integration
- User table creation
- Swagger setup

- Background startup scripts

Next:

- User Registration
- Password Hashing
- JWT Authentication
- Loan Application APIs